Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (Benzene)
Basis set   : external-CI
Charge   : 0
Multiplicity : 1
Total e-   : 6
n-body order : 5

GROUND STATE ENERGIES

E0 (Ha) : nan
E1 (Ha) : nan
Gap (Ha) : nan
HF (Ha) : nan

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 20

C_LB : coherence witness bound
C_gcg: Conjugate Gradient(Alg. A)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 0
Frozen e-   : 0
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 0.777858
<D> / site : 0.129643

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Body order : 5
Clusters   : 6

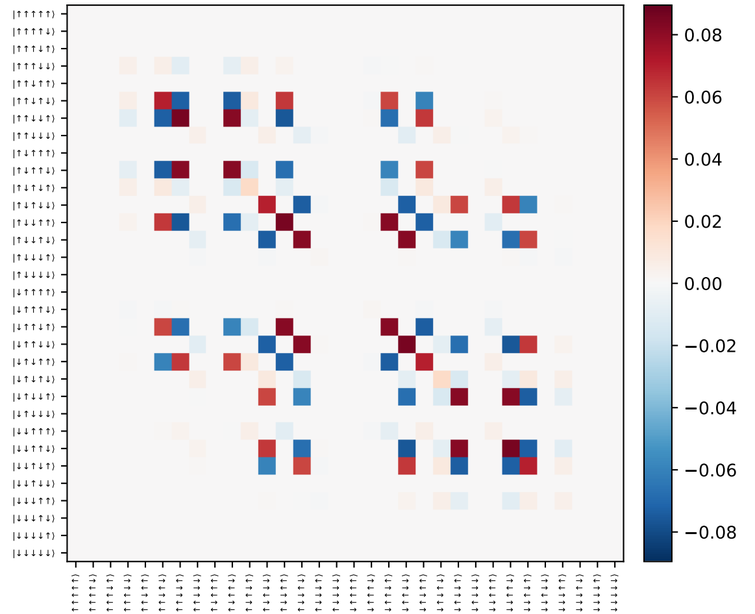
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Summary — 5-Body RDMs (Benzene, CAS(6,6))

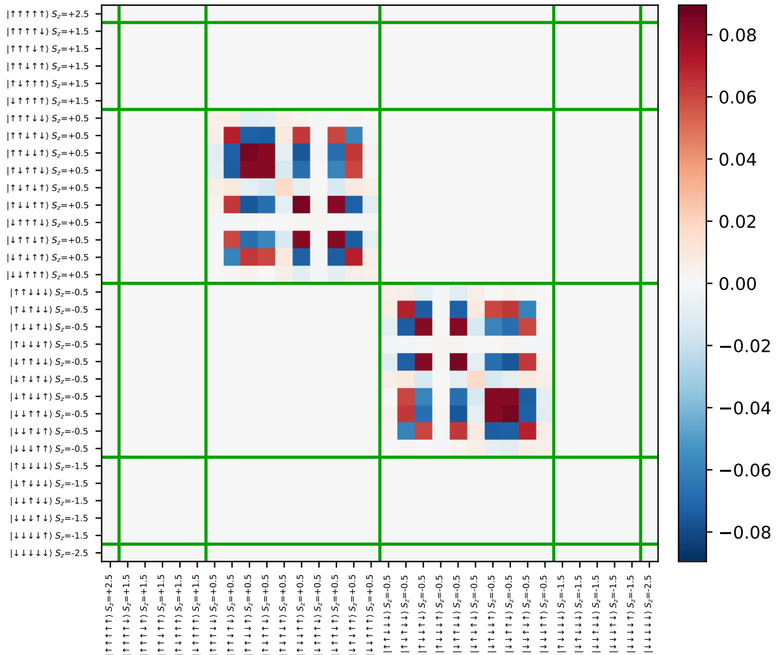
Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2,3,4	0.366792	1.213794	—	0.000000	0.614183	0.923341	MIXED
0,1,2,3,5	0.366792	1.213794	—	0.000000	0.612886	0.923341	MIXED
0,1,2,4,5	0.366792	1.213794	—	0.000000	0.614183	0.923341	MIXED
0,1,3,4,5	0.366792	1.213794	—	0.000000	0.614183	0.923341	MIXED
0,2,3,4,5	0.366792	1.213794	—	0.000000	0.614179	0.923341	MIXED
1,2,3,4,5	0.366792	1.213794	—	0.000000	0.613874	0.923341	MIXED

5-Body RDM (Sites 0,1,2,3,4) — same matrix in all three views

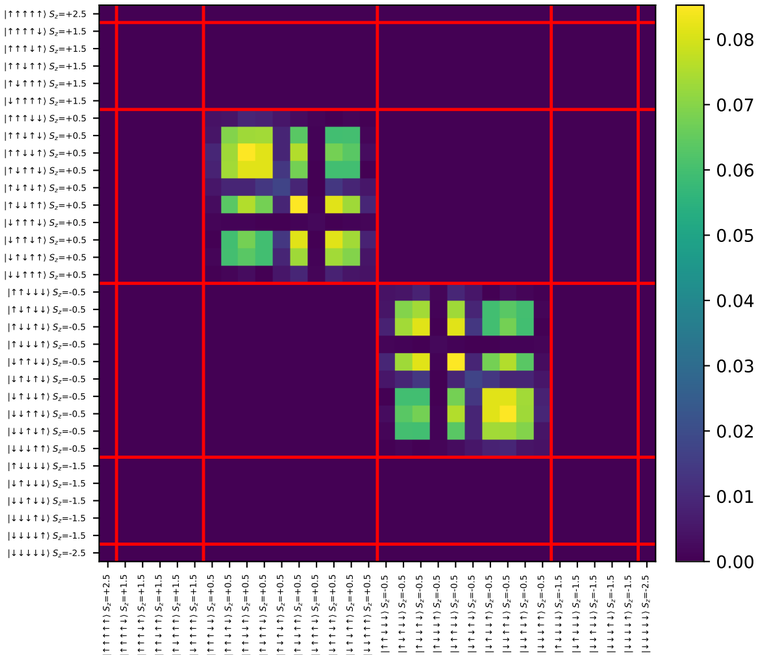
(1) Re(ρ) — index order



(2) Re(ρ) — S_z-ordered (original values)

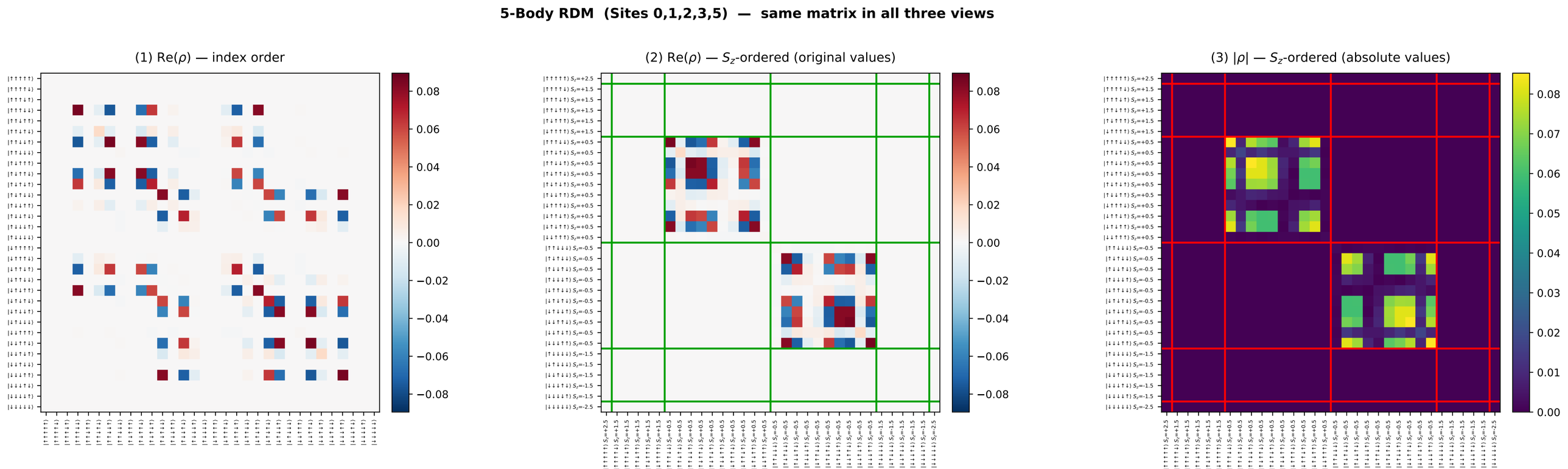


(3) $|\rho|$ — S_z-ordered (absolute values)



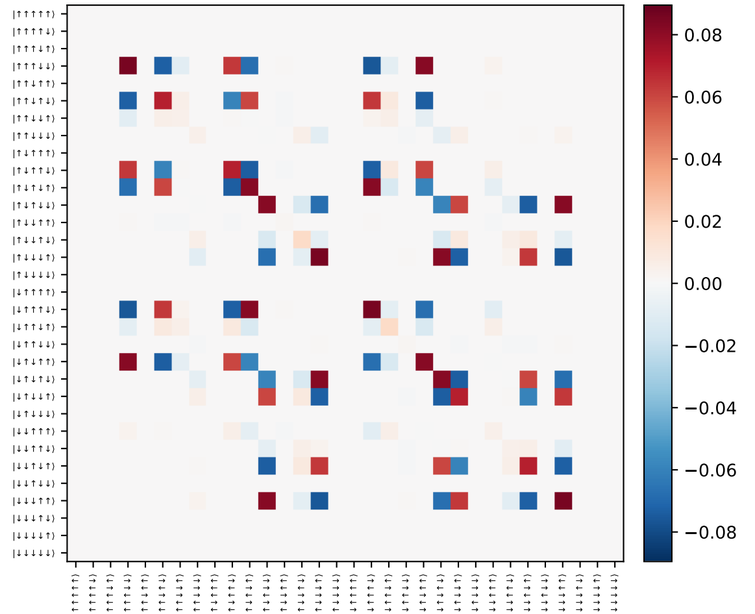
Trace=1.000000 Purity=0.366792 Entropy=1.213794 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.614183 ≤ C_{UB} = 0.923341 verdict: GME

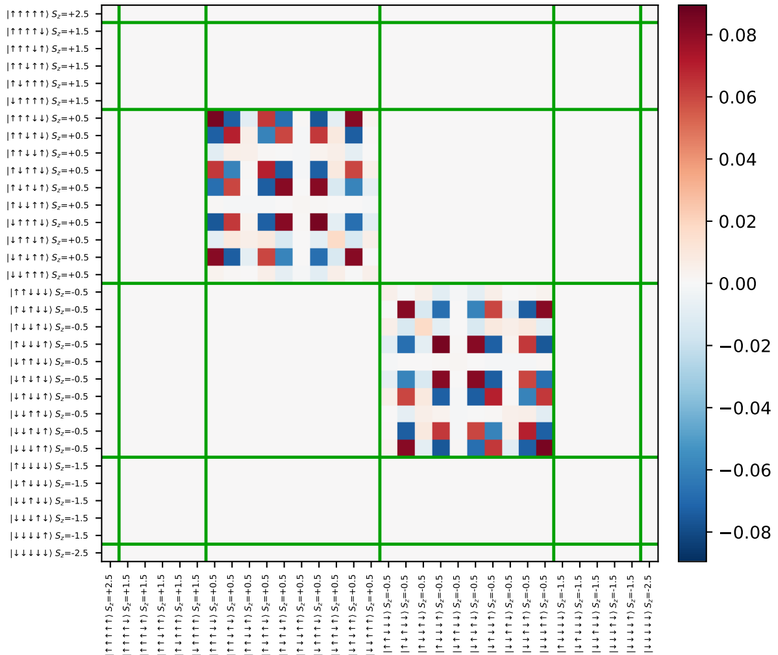

5-Body RDM (Sites 0,1,2,3,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.366792 Entropy=1.213794 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.612886 ≤ C_{UB} = 0.923341 verdict: GME

5-Body RDM (Sites 0,1,2,4,5) — same matrix in all three views

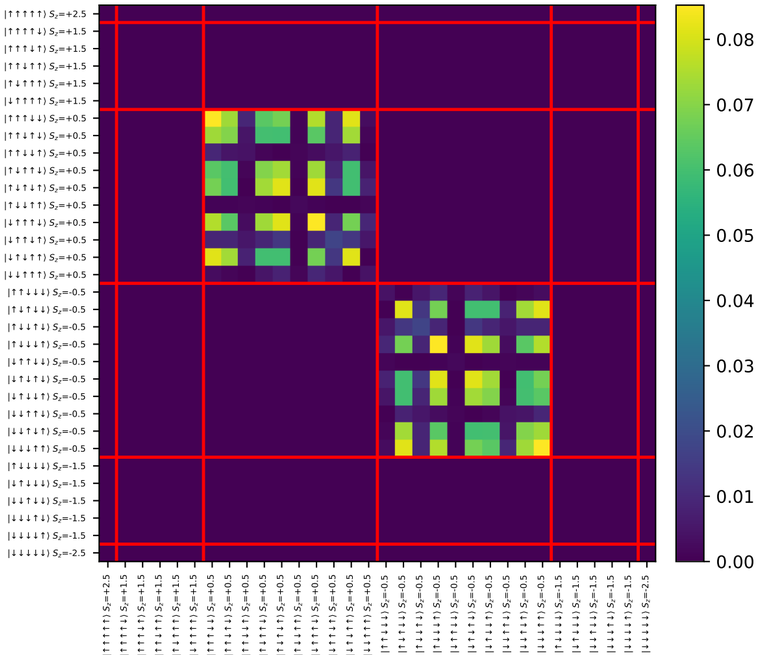
(1) Re(ρ) — index order



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(3) $|\rho|$ — S_z-ordered (absolute values)

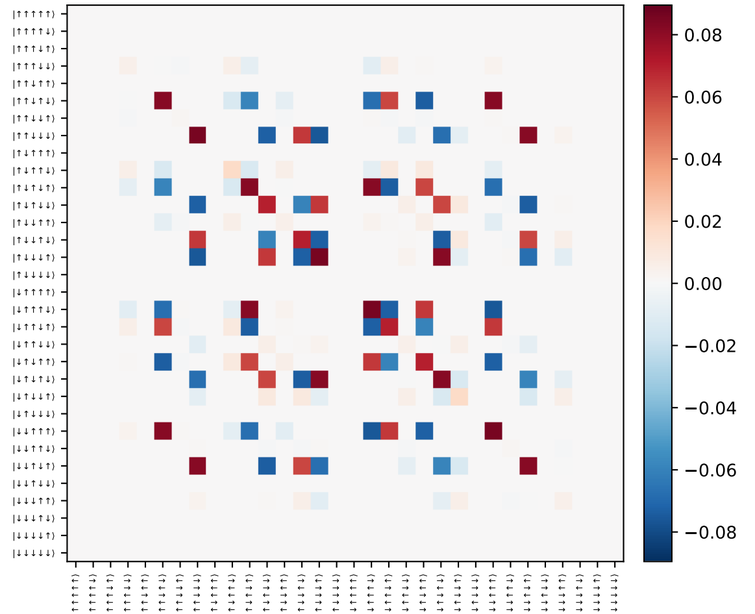


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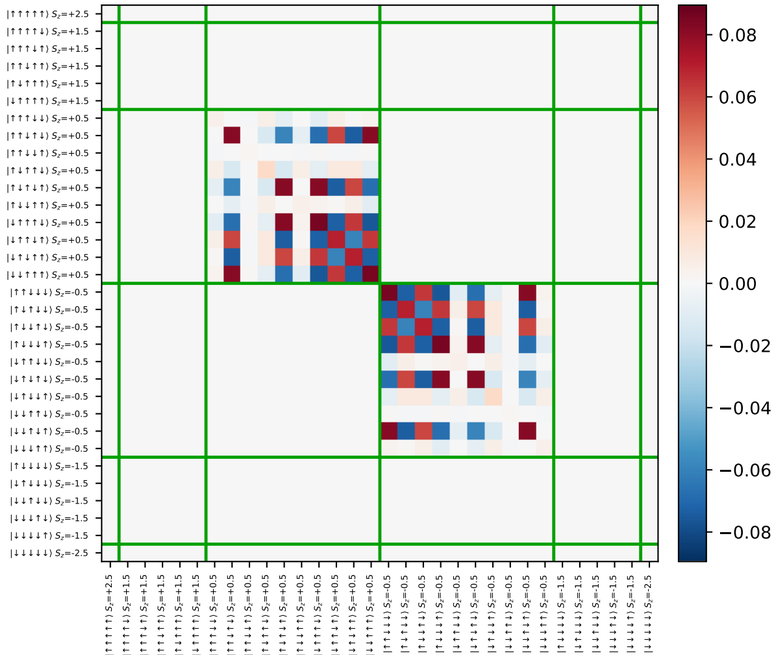
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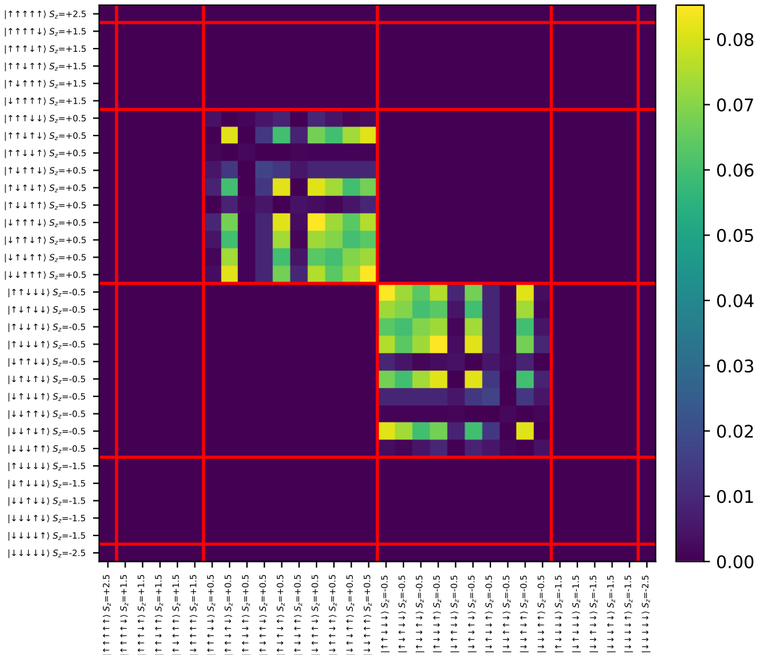
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(3) $|\rho|$ — S_z-ordered (absolute values)

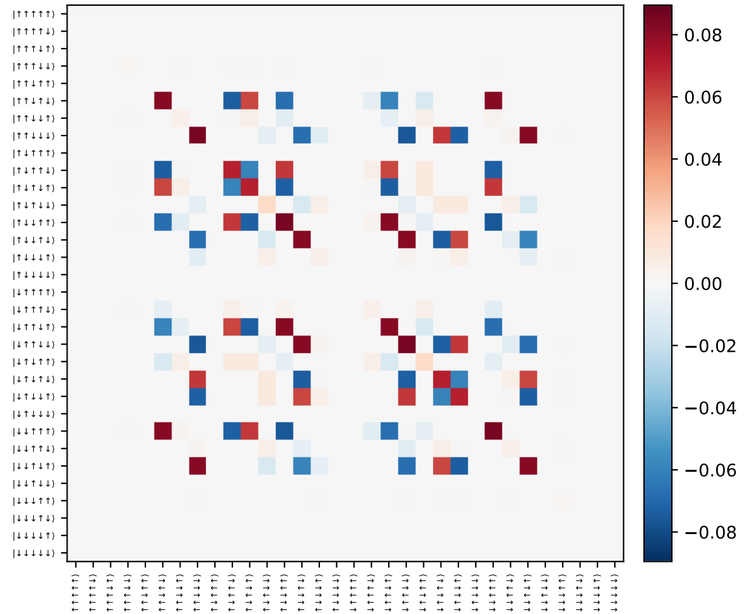


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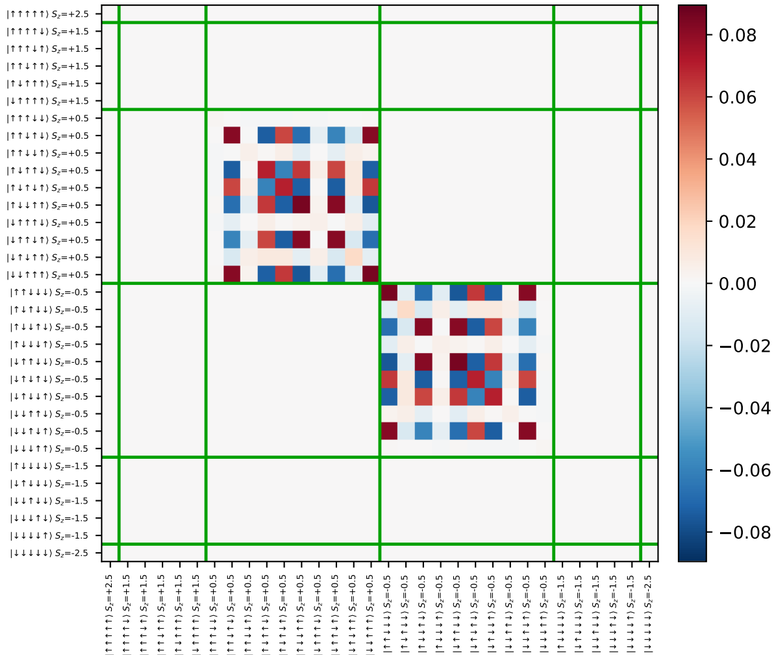
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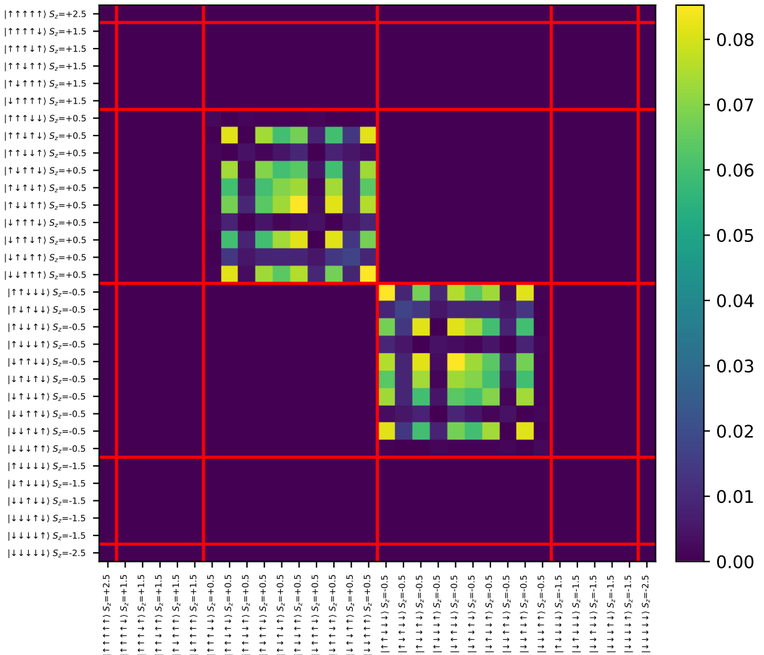
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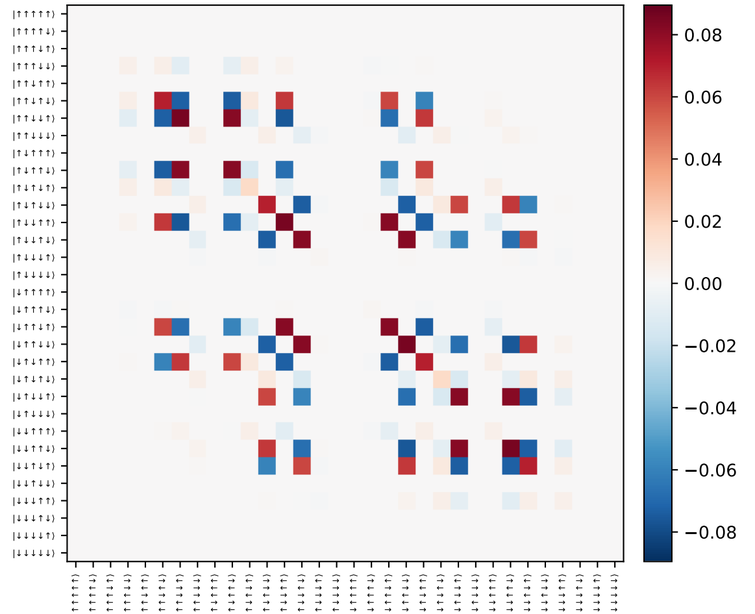


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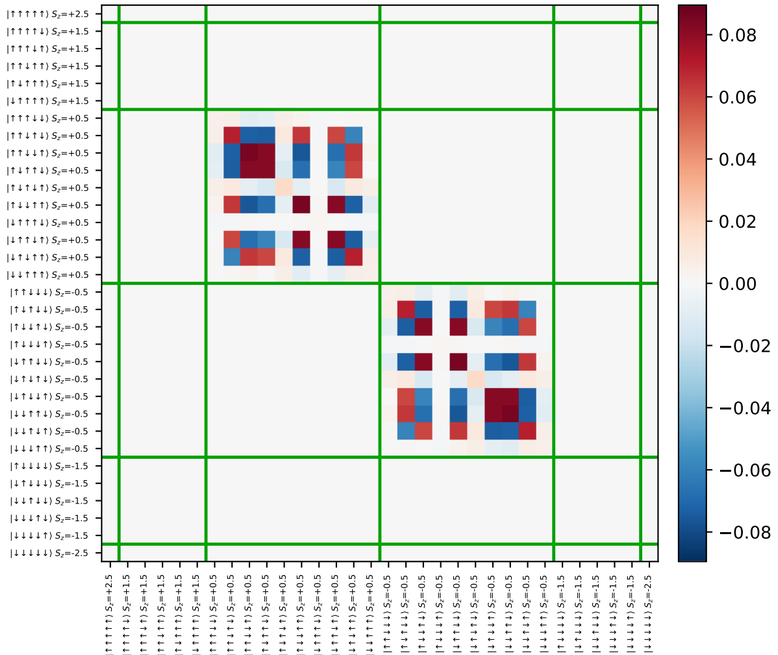
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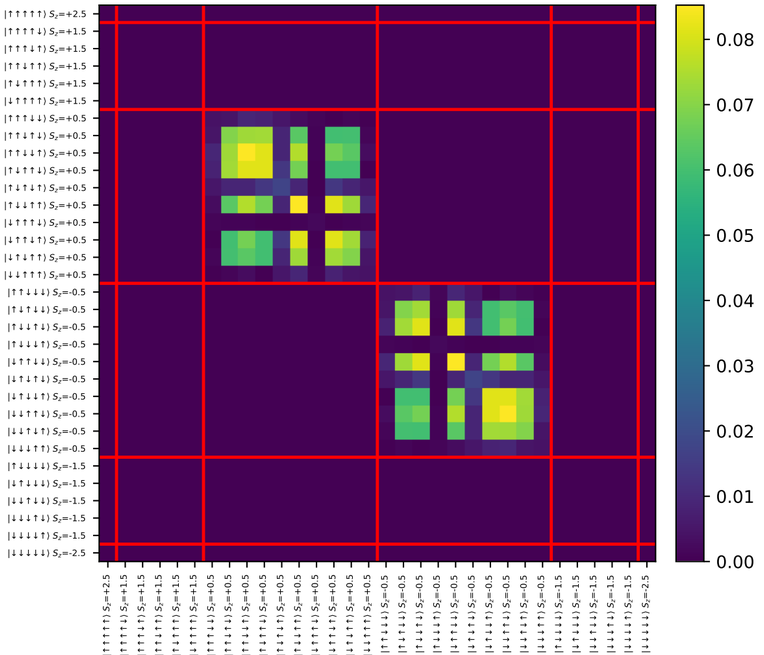
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